

Program : Diploma in Robotic Process Automation	
Course Code : 6301C	Course Title: Artificial Intelligence
Semester : 6	Credits: 4
Course Category: Program Elective	
Periods per week: 4 (L: 4 T: 0 P: 0)	Periods per semester: 60

Course Objectives:

- To understand fundamental principles of intelligent systems
- To study the characteristics of intelligent systems, knowledge representation schemes, logic and inference mechanisms.
- To understand the design of self learning systems along with some of their typical applications in the emerging scenario
- To understand the basic concepts in machine learning.

Course Outcomes:

On completion of the course student will be able to:

CO_n	Description	Duration (Hours)	Cognitive Level
CO1	Understand the fundamental concepts of intelligent systems and their architecture.	12	Understanding
CO2	Illustrate uninformed and informed search techniques for problem solving in intelligent systems.	19	Understanding
CO3	Solve Constraint Satisfaction Problems using search techniques.	13	Applying
CO4	Represent AI domain knowledge using logic systems and use inference techniques for reasoning in intelligent systems.	14	Applying
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO1	3	2	1				2
CO2	3	2	2				2
CO3	3	2	2				2
CO4	3	2	2				2

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Understand the fundamental concepts of intelligent systems and their architecture.		
M1.01	Introduction, What is Artificial Intelligence (AI)?	1	Understanding
M1.02	The foundations of AI	1	Understanding
M1.03	History of AI	1	Understanding
M1.04	Applications of AI	1	Understanding
M1.05	What is an Intelligent Agent?	1	Understanding
M1.06	Agents and Environments	1	Understanding
M1.07	The nature of Environments	2	Understanding
M1.08	The structure of Agents	4	Understanding
Contents: Introduction to AI – Introduction – What is Artificial Intelligence(AI) ? The Foundations of AI, History of AI,Applications of AI. Intelligent Agents – Agents and Environments, nature of Environments, Structure of Agents.			
CO2	Illustrate uninformed and informed search techniques for problem solving in intelligent systems.		

Module Outcomes	Description	Duration (Hours)	Cognitive Level
M2.01	Solving Problems by searching	1	Understanding.
M2.02	Problem solving Agents	1	Understanding.
M2.03	Sample problems – Toy problems	2	Applying
M2.04	8-puzzle problem	1	Applying
M2.05	8-queens problem	1	Applying
M2.06	Real world problems	3	Applying
M2.07	Uninformed search strategies: BFS,DFS,	2	Applying
M2.08	Uniform cost search	1	Applying
M2.09	Depth limited search.	1	Applying
M2.10	Informed search strategies	2	Applying
M2.11	Greedy best first search	1	Applying
M2.12	A* search.	2	Applying
M2.13	Heuristic functions.	1	Applying
	Series Test – I	1	
Contents: Solving Problems by searching : Solving Problems by searching, Problem solving Agents, Sample problems – Toy problems, 8-puzzle, 8-queens. Real world problems. Uninformed search strategies: BFS,DFS,Uniform cost search ,Depth limited search. Informed search strategies: Greedy best first search, A* search. Heuristic functions.			
CO3	Solve Constraint Satisfaction Problems using search techniques.		
M3.01	Adversarial search - Games	1	Understanding
M3.02	Optimal decisions in games	1	Understanding
M3.03	The Minimax algorithm	1	Applying
M3.04	Alpha-Beta pruning	2	Applying
M3.05	Defining CSP.	2	Applying

Module Outcomes	Description	Duration (Hours)	Cognitive Level
M3.06	Backtracking search for CSPs	2	Applying
M3.07	Structure of CSP	2	Applying
M3.08	Sample CSP problems	2	Applying
Contents: Adversarial search - Games, Optimal decisions in games, The Minimax algorithm, Alpha-Beta pruning. Constraint Satisfaction Problems – Defining CSP, Backtracking search for CSPs, Structure of CSP			
CO4	Represent AI domain knowledge using logic systems and use inference techniques for reasoning in intelligent systems.		
M4.01	Logical Agents	1	Understanding
M4.02	Knowledge based agents	2	Understanding
M4.03	Propositional Logic	2	Applying
M4.04	Propositional Theorem proving	1	
M4.05	Agents based on Propositional Logic.	1	
M4.06	First Order Predicate Logic	2	Applying
M4.07	Syntax and Semantics of First Order Logic	2	
M4.08	Using First Order Logic, Knowledge representation in First Order Logic.	2	Applying
	Series Test – II	1	
Contents: Logical Agents – Knowledge based agents, Logic, Propositional Logic, Propositional Theorem proving, Agents based on Propositional Logic. First Order Predicate Logic – Syntax and Semantics of First Order Logic, Using First Order Logic, Knowledge representation in First Order Logic.			

Text / Reference

T/R	Book Title/Author
T1	Stuart Russell and Peter Norvig. Artificial Intelligence: A Modern Approach, 3rd Edition. Prentice Hall.
T2	Nilsson N.J., Artificial Intelligence - A New Synthesis, Harcourt Asia Pvt. Ltd.